Happiest Health Digital Twin Challenge 2026

Sugar Twin India

A digital twin of each Type 2 diabetes patient that predicts glucose spikes and lows 2 hours before they happen

Team guptaaayush0908 · Maharaja Agrasen Institute of Technology

Reimagining and Reforming Healthcare in India Summit, Bengaluru

0.958

spike-alert AUROC on unseen patients

95%

of spikes flagged up to 2 h early

±13.8

mg/dL error, 1-hour forecast

Team guptaaayush0908

Maharaja Agrasen Institute of Technology (MAIT), Delhi



Aayush Gupta

Team leader

Maharaja Agrasen Institute of Technology



Dhruv Gandhi

Team member

Maharaja Agrasen Institute of Technology



Divyansh Singh

Team member

Maharaja Agrasen Institute of Technology

Diabetes is managed between visits, where no one is watching

101 M

people with diabetes in India

ICMR-INDIAB, 2023

Type 2 diabetes is the condition we chose: the most common lifestyle disease in India.

1

Harm happens between visits

Post-meal spikes from rice-heavy meals and late dinners, festivals, missed doses and night-time lows go unseen for months.

2

A CGM only shows the past

By the time an alarm sounds, the spike or the low is already happening.

3

Too much data for a doctor

288 CGM readings a day per patient, plus a smartwatch and a 3-year health record, for hundreds of patients.

A twin of each patient that warns early and tests advice first

Fuses the electronic health record (static) with CGM and smartwatch data (dynamic)

1

Triage

Every patient sorted by risk right now: urgent (a low or above 250 likely) and watch (a spike likely).

2

Early warning

Spike above 180, above 250 or a low below 70 predicted up to 2 hours ahead, with plain-language reasons.

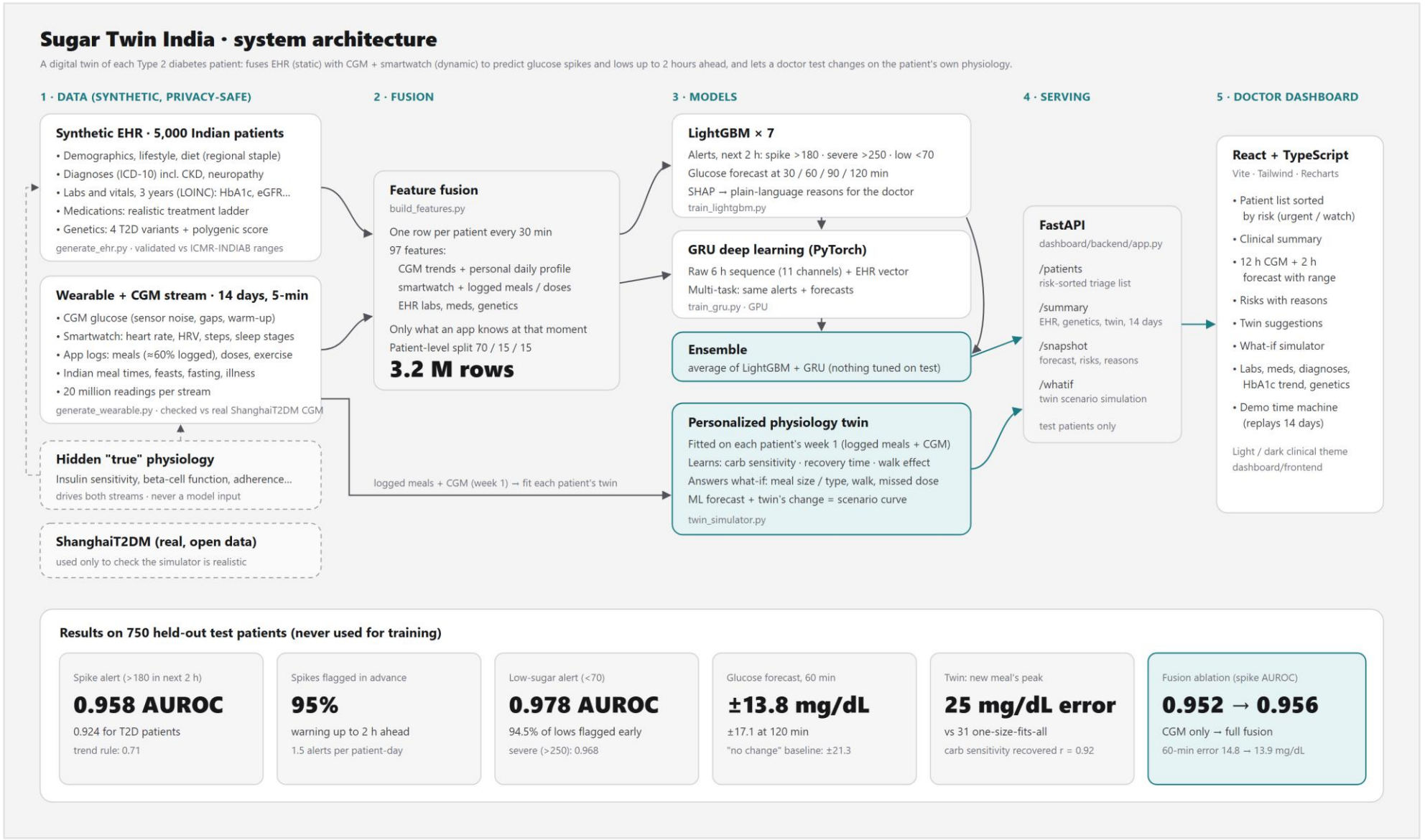
3

Advice tested on the twin

The patient's own physiology simulates a smaller meal, a walk or a missed dose before the doctor advises it.

Doctor's question answered in seconds: who needs me now, and what should I tell them?

How it works


Sugar Twin India · system architecture
A digital twin of each Type 2 diabetes patient: fuses EHR (static) with CGM + smartwatch (dynamic) to predict glucose spikes and lows up to 2 hours ahead, and lets a doctor test changes on the patient's own physiology.
1 · DATA (SYNTHETIC, PRIVACY-SAFE)
Synthetic EHR · 5,000 Indian patients
• Demographics, lifestyle, diet (regional staple)
• Diagnoses (ICD-10) incl. CKD, neuropathy
• Labs and vitals, 3 years (LOINC): HbA1c, eGFR...
• Medications: realistic treatment ladder
• Genetics: 4 T2D variants + polygenic score
generate_ehr.py · validated vs ICMR-INDIAB ranges
Wearable + CGM stream · 14 days, 5-min
• CGM glucose (sensor noise, gaps, warm-up)
• Smartwatch: heart rate, HRV, steps, sleep stages
• App logs: meals (≈60% logged), doses, exercise
• Indian meal times, feasts, fasting, illness
• 20 million readings per stream
generate_wearable.py · checked vs real ShanghaiT2DM CGM
Hidden "true" physiology
Insulin sensitivity, beta-cell function, adherence...
drives both streams · never a model input
ShanghaiT2DM (real, open data)
used only to check the simulator is realistic
2 · FUSION
Feature fusion
build_features.py
One row per patient every 30 min
97 features:
CGM trends + personal daily profile
smartwatch + logged meals / doses
EHR labs, meds, genetics
Only what an app knows at that moment
Patient-level split 70 / 15 / 15
3.2 M rows
logged meals + CGM (week 1) → fit each patient's twin
3 · MODELS
LightGBM × 7
Alerts, next 2 h: spike >180 · severe >250 · low <70
Glucose forecast at 30 / 60 / 90 / 120 min
SHAP → plain-language reasons for the doctor
train_lightgbm.py
GRU deep learning (PyTorch)
Raw 6 h sequence (11 channels) + EHR vector
Multi-task: same alerts + forecasts
train_gru.py · GPU
Ensemble
average of LightGBM + GRU (nothing tuned on test)
Personalized physiology twin
Fitted on each patient's week 1 (logged meals + CGM)
Learns: carb sensitivity · recovery time · walk effect
Answers what-if: meal size / type, walk, missed dose
ML forecast + twin's change = scenario curve
twin_simulator.py
4 · SERVING
FastAPI
dashboard/backend/app.py
/patients
risk-sorted triage list
/summary
EHR, genetics, twin, 14 days
/snapshot
forecast, risks, reasons
/whatif
twin scenario simulation
test patients only
5 · DOCTOR DASHBOARD
React + TypeScript
Vite · Tailwind · Recharts
• Patient list sorted by risk (urgent / watch)
• Clinical summary
• 12 h CGM + 2 h forecast with range
• Risks with reasons
• Twin suggestions
• What-if simulator
• Labs, meds, diagnoses, HbA1c trend, genetics
• Demo time machine (replays 14 days)
Light / dark clinical theme
dashboard/frontend
Results on 750 held-out test patients (never used for training)
Spike alert (>180 in next 2 h)
0.958 AUROC
0.924 for T2D patients
trend rule: 0.71
Spikes flagged in advance
95%
warning up to 2 h ahead
1.5 alerts per patient-day
Low-sugar alert (<70)
0.978 AUROC
94.5% of lows flagged early
severe (>250): 0.968
Glucose forecast, 60 min
±13.8 mg/dL
±17.1 at 120 min
"no change" baseline: ±21.3
Twin: new meal's peak
25 mg/dL error
vs 31 one-size-fits-all
carb sensitivity recovered r = 0.92
Fusion ablation (spike AUROC)
0.952 → 0.956
CGM only → full fusion
60-min error 14.8 → 13.9 mg/dL

Two data streams, one virtual body

All synthetic: no real patient records (DPDP Act and HIPAA safe)

Static: synthetic EHR · 5,000 Indian patients

- Demographics, lifestyle, regional diet
- Diagnoses coded in ICD-10
- 3 years of labs and vitals (LOINC)
- Medicines on a realistic treatment ladder
- Genetic markers: 4 T2D variants + polygenic score

Dynamic: wearables · 14 days, every 5 min

- CGM glucose with sensor noise and gaps
- Heart rate, HRV, steps, sleep stages
- Logged meals (about 60%), doses, exercise
- Indian meal times, feasts, fasting, illness
- 20 million readings per stream

r = 0.93

CGM-estimated vs lab HbA1c: both streams describe one body

82% vs 84%

time in range: simulated vs real ShanghaiT2DM CGM

0

hidden physiology values given to the models

Two kinds of twin: one predicts, one simulates

1 LightGBM × 7

Alerts for the next 2 h (spike above 180, above 250, low below 70) and forecasts at 30 to 120 min. SHAP gives plain-language reasons.

2 GRU deep learning

Reads the raw 6-hour sequence (11 channels) plus the EHR vector; multi-task alerts and forecasts.

3 Ensemble

Plain average of LightGBM and GRU: nothing tuned on test patients. Feeds the triage list and the forecast chart.

4 Personal physiology twin

Learns each patient's carb sensitivity, recovery time and walking effect from week 1; simulates meals, walks and missed doses.

It sees spikes and lows coming

750 held-out test patients, never used for training

0.958

spike-alert AUROC (0.924 for T2D) vs 0.71 for a trend rule

95%

of spikes flagged up to 2 hours before they happen

0.978

low-sugar AUROC; 94.5% of lows flagged early

±13.8

mg/dL error at 60 min vs ±21.3 assuming no change

Glucose forecast	30 min	60 min	90 min	120 min
Sugar Twin error (mg/dL)	10.6	13.8	15.9	17.1
'Glucose stays the same' baseline	14.1	21.3	27.3	31.7

Spike alerts at 80% sensitivity: precision 0.75, about 1.5 alerts per patient-day. Clarke error grid zones A + B above 99% at every horizon.

Fusing the streams helps, and personal beats average

Inputs	Spike AUROC	T2D	60-min error
CGM only	0.952	0.915	14.8
CGM + smartwatch + logs	0.955	0.919	14.1
CGM + EHR	0.953	0.917	14.6
Full fusion	0.956	0.921	13.9

Each stream adds information; full fusion is best for spikes and forecasts. Low-sugar prediction comes mostly from the CGM itself.

Personal twin

25 mg/dL

error predicting a new meal's peak

vs 31 mg/dL one-size-fits-all

Carb sensitivity learned from week 1 matches the hidden truth: r = 0.92

Tested the way it will be used

OK	**Day 1, no history yet**	**spike AUROC 0.955**
OK	**Patient never logs meals**	**0.955**
OK	**No smartwatch, CGM only**	**0.956**
OK	**10% noisier sensor**	**0.942**

19 of 24 clinical scenario checks pass

Feasts, missed doses, illness, fasting on sulfonylureas, night-time lows, healthy people after a big meal.

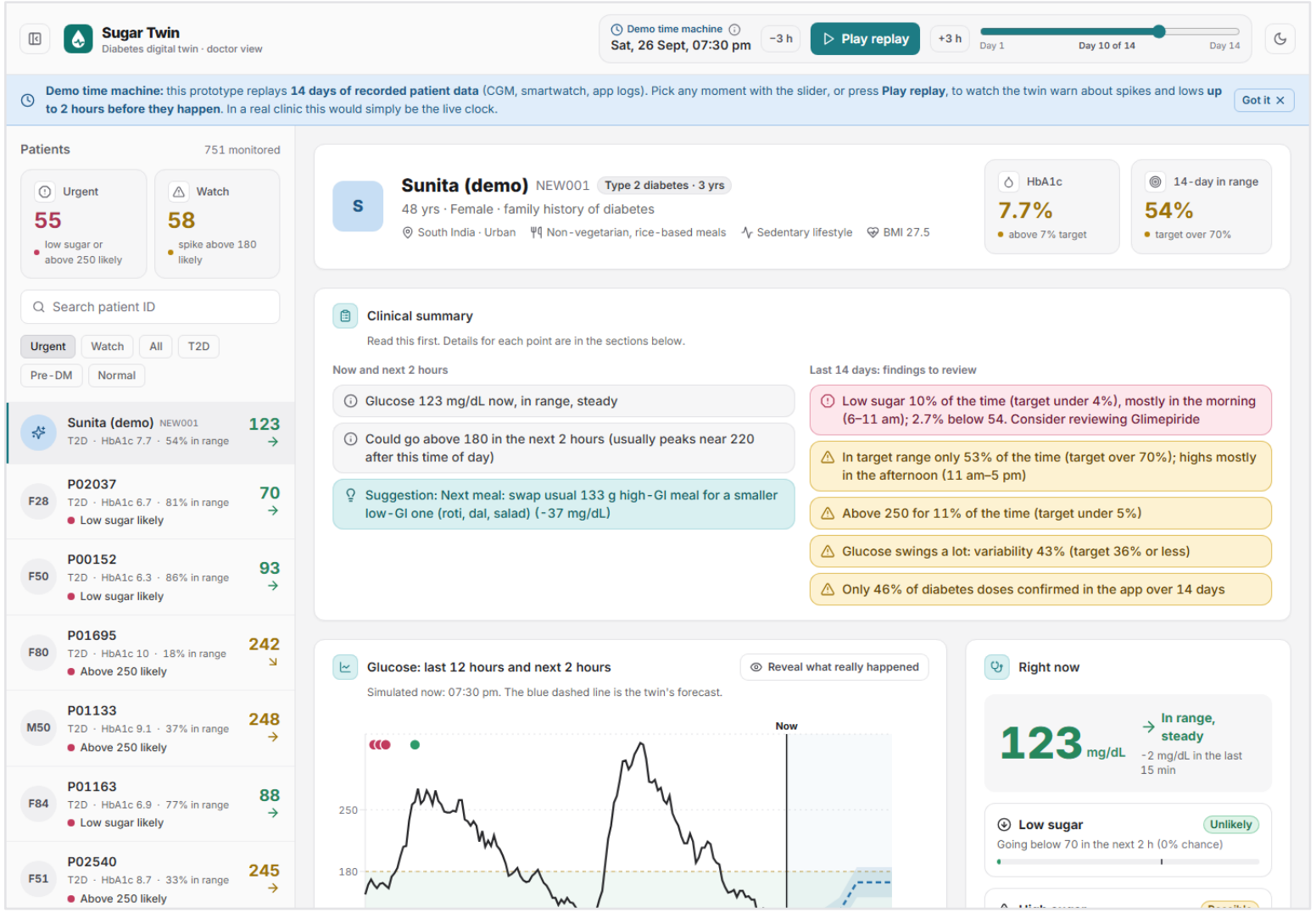
The 5 that fail are what-if checks on the ML models (for example, a walk should lower the forecast): they learn correlation, not cause and effect.

That is why what-if questions go to the physiology twin.

Brand-new patient 'Sunita', created outside the dataset

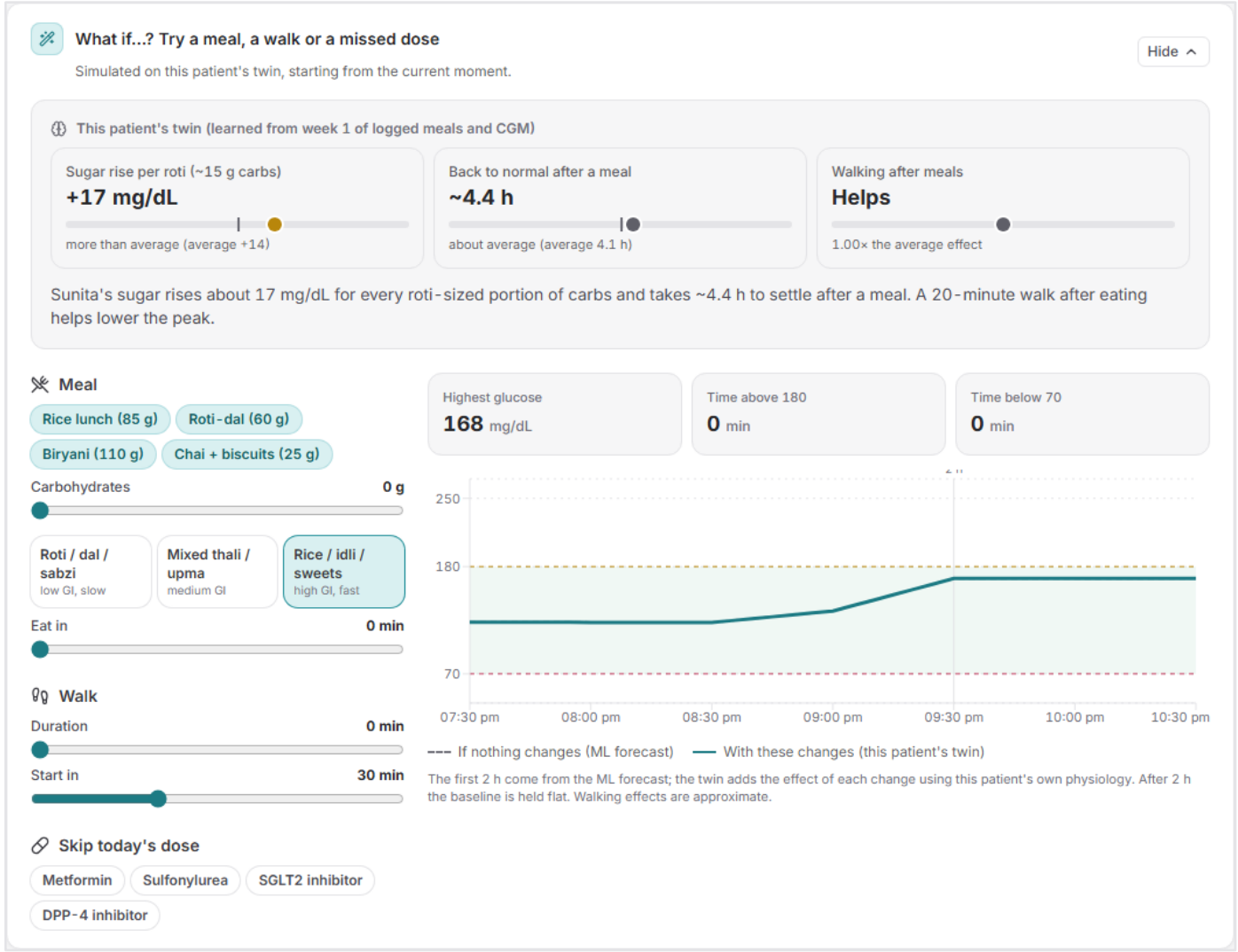
All 48 of her spikes and 18 lows flagged in advance, about 100 minutes early on average.

What the doctor sees



- Patients sorted by risk: urgent and watch
- Clinical summary in plain language
- 12-hour CGM plus 2-hour forecast with likely range
- Risks with reasons from the model
- Twin suggestions with their simulated effect
- Dropdowns: what-if, health record, genetics, accuracy
- Demo time machine replays 14 days

Advice tested on the patient's own twin



The twin learns three things

How much 10 g of carbs raises glucose, how fast it recovers, and how much a walk helps, from week 1 of logged meals and CGM.

The doctor tries a change

Meal size and type, a walk after eating, or a missed dose: the curve updates against 'if nothing changes'.

Example: Sunita

Swapping her usual 133 g high-GI meal for a smaller low-GI one lowers the predicted peak by about 37 mg/dL.

Honest limits and what comes next

Limitations

- All data is synthetic; accuracy on real patients will be lower
- The twin shares equations with the simulator, so its recovery of physiology is an upper bound
- Low-sugar alerts are sensitive but give false alarms (about 3 in 4)
- Warning time is measured on a 30-minute grid

Next steps

- Validate on real, consented CGM data with a partner hospital
- Re-fit each twin continuously as new meals are logged
- Patient app with an 'I'm about to eat' button (spike AUROC 0.975 with a meal plan)
- Same EHR + wearable fusion for hypertension and cardiovascular risk

Sugar Twin India

Who needs me now, and what should I tell them?

A digital twin that answers the doctor's question in seconds, two hours before the event.

github.com/Aayushg0908/sugar-twin-india

Team guptaaayush0908 · Maharaja Agrasen Institute of Technology · MIT License

All patient data is synthetic. Decision-support prototype, not a medical device.